

RESEARCH ARTICLE

Research on Gait Recognition Based on GaitSet and Multimodal Fusion

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ABSTRACT With the continuous technological progress, especially the development in biometrics, gait recognition has shown broad application prospects in healthcare (e.g., health monitoring), security (e.g., assisted identity verification), and human-computer interaction. However, individual differences, such as changes in physical condition, and environmental variability, such as differences in lighting, can impact its accuracy. Based on the information derived from the gait contour sequence during walking (such as temporal and spatial information), this study proposes an improved gait recognition method based on the GaitSet model, which improves video-based gait recognition performance by combining gait energy images and silhouette images to form a multimodal representation. The experimental results showed a significant performance improvement compared with the original model, especially in subjects with bags. Large-sample training experiment results based on the CASIA-B database indicated that the recognition rates in the Normal (NM), Bag (BG), and Coat (CL) states were 95.8%, 89.3%, and 72.5%, respectively, and that in the CL state achieved a significant improvement of 3.3%.

INDEX TERMS Deep learning, gait recognition, interpolation, multimodal attention mechanism.

I. INTRODUCTION

Gait recognition is a biometric technology that uses an individual's walking pattern as a means of identification [1]. Compared with other biometrics, e.g., face, fingerprint, and iris, gait patterns can be captured from a distance in uncontrolled settings without requiring any physical contact and the cooperation of the subject that has to be identified [2]. With continuous social and technological advancements, gait recognition has shown wide application prospects in security and surveillance, healthcare and disease diagnosis, and human-computer interaction [3], [4]. However, gait recognition often suffers from multiple exterior factors in real scenes, such as carrying conditions, wearing overcoats, and diverse viewing angles [5].

Research on gait recognition generally includes model-based and non-model-based methods. Model-based gait

recognition methods mainly model the structure and movement of the human body. Lee and Grimson [6] first proposed an elliptical model for gait recognition, and Yoo et al. [7] proposed a stick figure model in the same year. To address issues in practical applications such as perspective changes and complex scenarios with or without occlusion, Urtasun and Fua [8] proposed a 3D (stick figure) model to calculate the gait pendulum inclination angle based on temporal data. Castro et al. [9] propose AttenGait, a novel model for gait recognition equipped with trainable attention mechanisms that automatically discover interesting areas of the input data. AttenGait can be used with any informative modalities, such as optical flow, obtaining state-of-the-art results thanks to the richer information contained in those modalities.

Non-model-based gait recognition methods achieve identification mainly by extracting human gait features such as the contour and silhouette from the original image [10]. Han and Bhanu [11] proposed the gait energy image to characterize the gait characteristics of the human body for

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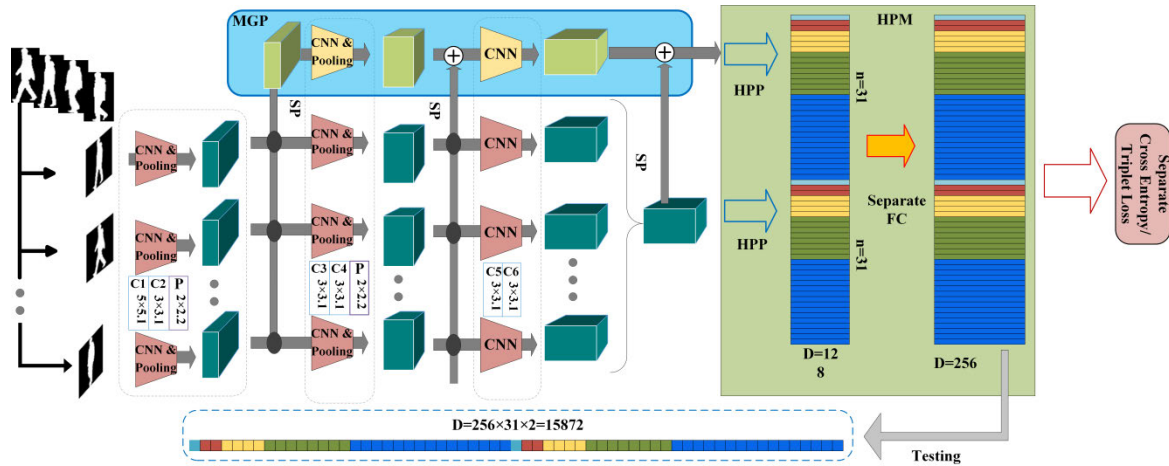


FIGURE 1. Framework diagram of the original GaitSet model.

individual recognition, which uses the original gait contour image to calculate the real GEI. Wang and Xu [12] employed a gait entropy image (GENI) to represent the gait image, which encoded the randomness of the contour image pixel values in the full gait cycle within a single image. As GENI mainly reflects the motion information, it has a certain robustness to appearance changes. Qiu et al. [13] propose a novel network structure GaitNet, which treats gait as a set of disordered sequences, compares different sequences to extract features, and introduces attention to gait recognition, thereby achieving a good recognition score under different angles and carrying situations of different clothes/bags. Chao et al. [14] proposed the GaitSet model, which models human gait through pose estimation and motion feature extraction. GaitSet has shown strong robustness and can reach satisfactory accuracy in relatively small test samples.

This paper focuses on a gait recognition method based on GaitSet and improves and optimizes the pre-processing and channel aspects to enhance the recognition rate. Overall, the contributions of this study can be summarized in the following three points: (1) The identification process of GaitSet is presented, and the pre-processing method is analyzed; (2) GEI and silhouette image are combined, and a multimodal attention mechanism is employed to extract key information, thus improving the gait recognition accuracy; (3) The performance of the improved model is verified by experiments on the CASIA-B database. The experimental results showed significantly improved recognition rates in different states (normal, NM; bag, BG; and coat, CL), especially in the CL state, where the recognition rate was increased by 3.3%. Compared to the original GaitSet model, the overall recognition rate is improved by 1.4%.

The current excellent models, such as GaitNet [13] and Gaitpart [15], have demonstrated remarkable performance in gait recognition, but require a lot of computational resources, which can be a limiting factor when it comes to actual deployment. In addition, when the human body is partially obscured, such as carrying objects or wearing

heavy clothing, the accuracy of gait recognition may be reduced. The multimodal approach proposed in this study has improved in handling occlusion, but this problem is still not completely solved. In response to the limitations of existing technologies, we are dedicated to developing more efficient and accurate recognition methods. Through continuous optimization and innovation, this method can enhance recognition accuracy, optimize algorithm structure, and utilize high-performance computing technology to achieve rapid and accurate gait recognition, thereby satisfying the demands of practical applications in smart cities and other fields.

II. GAITSET MODEL AND ITS IMPROVEMENTS

A. GAITSET MODEL WORKFLOW

The improvements in this study are based on the original Gaitset model, and the overall pipeline is illustrated in FIGURE 1. SP represents set pooling. MGP represents a multilayer global pipeline. HPP represents horizontal pyramid pooling. HPM is a structure called horizontal pyramid mapping. Trapezoids represent convolution and pooling blocks, and those in the same column have the same configurations, as shown by the rectangles with capital letters [14]. The specific process is as follows:

Step 1, the database is pre-processed, whose input is a set of gait contour images, which are cropped and aligned according to the size of the human body.

Step 2, each contour image is input into the convolutional neural network to extract its feature parameters, which are then aggregated into an advanced feature map through the Set Pooling (SP) operation, thereby better preserving the temporal and spatial characteristics of the image.

Step 3, the GEI multimodal attention mechanism is introduced to extract more valuable information, which accurately screens the key information in the feature map through deep feature extraction using two fully connected layers.

Step 4, GaitSet employs an HPM structure to map features to a more recognizable space through global maximum

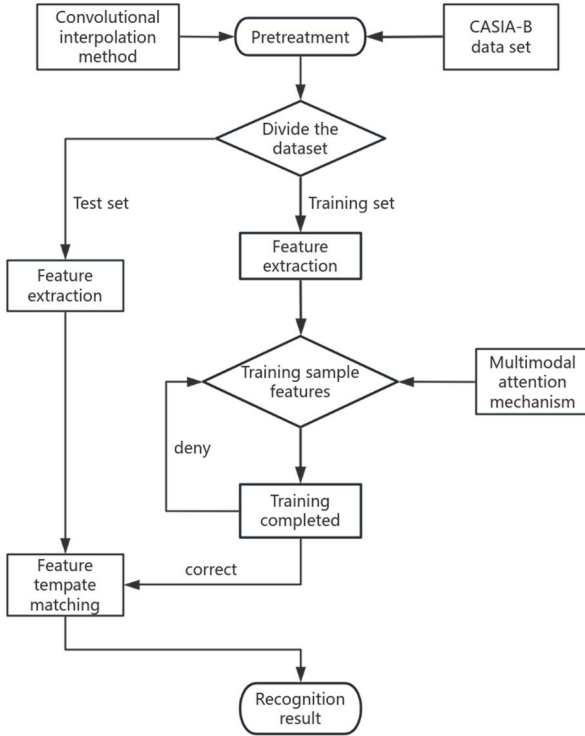


FIGURE 2. GaitSet gait recognition flowchart.

pooling and global average pooling, thus obtaining the final representation [16].

Step 5, the designed network model is optimized in the training phase through gradient descent and backpropagation and using the batch all triplet loss function, granting it the discriminative ability for pedestrian gait.

Step 6, Rank1 recognition rate judgment: This study uses a training set to train the GaitSet network model in FIGURE 1 and optimize the weights. Then, the samples to be tested in the test set are sent to the trained network model to obtain the gait features, which are then compared with target samples in the test set. The correct recognition rate is determined based on the Euclidean distance [17].

FIGURE 2 illustrates the gait recognition process conducted by the GaitSet model. This study primarily focuses on enhancing the pre-processing and training model components based on the original author's work.

The innovation of our work lies in: 1. Exploring the unexplored territory of integrating a novel fusion technique with multimodal biometric data in gait recognition research. 2. Proposing an enhanced gait set algorithm to address challenges encountered in gait recognition, such as variations in lighting and occlusion.

B. PRE-PROCESSING METHOD ANALYSIS

Normalization is a very important process in database image pre-processing as it directly affects the extraction of contour features by network models [18]. This process eliminates differences in brightness, contrast, etc., between different

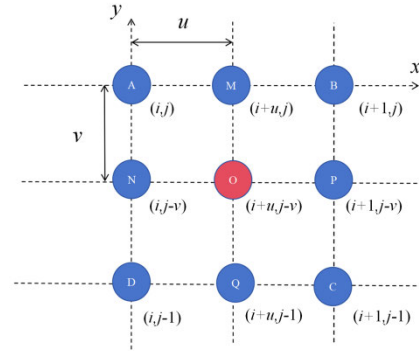


FIGURE 3. Schematic diagram of the bilinear interpolation method.

images, thereby facilitating fast model convergence and improving its performance.

Bilinear interpolation and cubic convolution interpolation are commonly used interpolation algorithms in image processing, which resample pixels during operations such as image scaling or rotation and estimate the value of unknown pixels [19]. Bilinear interpolation operates based on the four-pixel points around the pixel to be interpolated and performs linear interpolation once in both directions (horizontal and vertical), as shown in FIGURE 3.

A , B , C , and D are points actually present in the original image. Let $dx = u$, $dy = v$, the coordinates of point $M(i + u, j)$ are calculated by interpolating points $A(i, j)$ and $B(i + 1, j)$. The coordinates of $N(i, j - v)$, $P(i + 1, j - v)$, and $Q(i + u, j - 1)$ can be calculated in the same way, and the coordinates of the target point $O(i + u, j - v)$ can be obtained through linear estimation. The pixel value for each point in FIGURE 3 is denoted as $f(x, y)$, and x and y represent the coordinate values of the corresponding point. Its coordinates in the low-definition image determine the pixel values of four points in the peripheral neighborhood, which are $f(i, j)$, $f(i, j + 1)$, $f(i + 1, j)$, $f(i + 1, j + 1)$. Then, the pixel value of point O is expressed in Equation (1):

$$\begin{aligned} f(i + u, j - v) &= (1 - u) \cdot (1 - v) \cdot f(i, j) + (1 - v) \cdot u \cdot f(i, j + 1) \\ &\quad + v \cdot (1 - u) \cdot f(i + 1, j) + u \cdot v \cdot f(i + 1, j + 1) \end{aligned} \quad (1)$$

Equation (1) indicates that the bilinear interpolation method involves a relatively small amount of computation, with a low algorithm complexity, and overcomes the grayscale value discontinuity of the neighbor interpolation method. However, the bilinear interpolation method has the characteristics of a low-pass filter, causing losses of the high-frequency part of the scaled image, i.e., the image may be smoother and lose some details.

The cubic convolution interpolation method operates based on the adjacent 16 pixels (4×4 neighborhood) of the interpolation point and uses the segmented cubic polynomial function to weigh the pixels of the interpolation point [20], [21]. As shown in FIGURE 4, the pixel values of each point to be interpolated in the reconstructed image are calculated

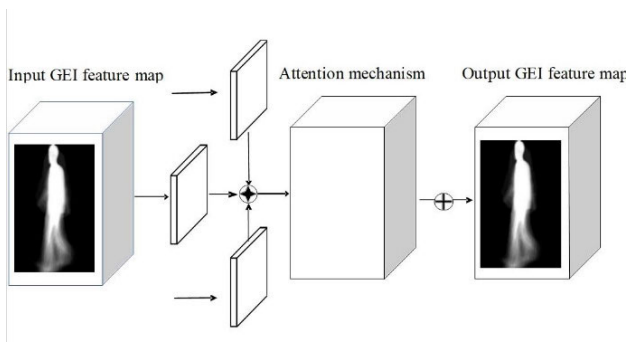


FIGURE 4. Schematic diagram of the cubic convolution interpolation method.

based on each 16-point region in the low-resolution image. Let point P be the center of the 4×4 region, and the remaining orange dots are used to indicate the points to be involved in the convolution operation. Based on $P(i, j)$, the coordinates of the remaining points can be determined. The point to be interpolated in the reconstructed image maps to the low-definition image at points $A1$ and $A2$. $A1$ and $A2$ falling within 0.5 lengths above, below, left, and right of point P are used as the criteria for using point P as the center point of the 16 points around the area, thereby determining the position and pixel value of all orange dots around it. The interpolation kernel function u is defined as follows:

$$u(s) = \begin{cases} a_1 |s|^3 + b_1 |s|^2 c_1 |s| + d_1 & 0 < |s| < 1 \\ a_2 |s|^3 + b_2 |s|^2 c_2 |s| + d_2 & 1 < |s| < 2 \\ 0 & 2 < |s| \end{cases} \quad (2)$$

To render function $u(s)$ and its derivative $u'(s)$ continuous, there has to be $u(0) = 1$, $u(1) = u(2) = 0$.

Equation (2) is an indefinite equation, and a_2 is selected as a parameter variable. Then, the constraint of Taylor expansion yields $a_2 = -0.5$, and Equation (2) is solved as:

$$u(s) = \begin{cases} \frac{3}{2} |s|^3 - \frac{5}{2} |s|^2 + 1 & 0 < |s| < 1 \\ -\frac{1}{2} |s|^3 + \frac{5}{2} |s|^2 - 4 |s| + 2 & 1 < |s| < 2 \\ 0 & 2 < |s| \end{cases} \quad (3)$$

The pixel coordinates of the image reconstructed through cubic convolution interpolation are (x, y) , and the coordinates mapped to the low-definition image are calculated as:

$$\begin{cases} i = \left(x + \frac{1}{2}\right) \cdot h^{-1} - \frac{1}{2} \\ j = \left(y + \frac{1}{2}\right) \cdot h^{-1} - \frac{1}{2} \end{cases} \quad (4)$$

According to Equation (3), the coefficients of the 4 points in the xy direction are $u(x1)$, $u(x2)$, $u(x3)$, $u(x4)$, $u(y1)$, $u(y2)$, $u(y3)$, and $u(y4)$. In FIGURE 3 and FIGURE 4, the X-axis and Y-axis represent the horizontal and vertical directions of the image without involving specific physical dimensions. By using $I(x, y)$ to represent the pixel value of a point in the image and $I_{new}(x, y)$ to represent the pixel value of a point in the reconstructed image, the calculation of the reconstructed pixels is shown in Equation (5), as shown at the bottom of page 6.

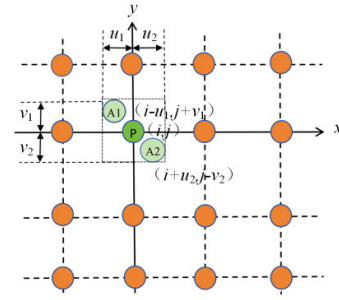


FIGURE 5. Normalized gait contour diagram.

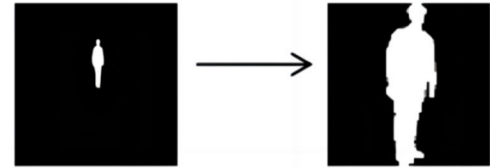


FIGURE 6. Multimodal attention mechanism.

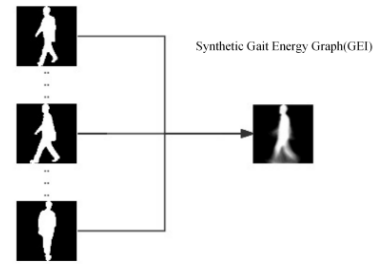


FIGURE 7. Gait energy image.

The effect of sample normalization in the CASIA-B database is illustrated in FIGURE 5, where the pre-processed silhouette image is magnified to provide more detailed features of the image and clearly show the outline integrity of the figure, including the details of the limbs, trunk, and other parts. Cubic convolution interpolation calculates the interpolated pixel value through the cubic polynomial and the 4×4 pixel matrix, thereby producing smoother edges while scaling the image, which in turn reduces aliasing. In this way, higher-quality images with reduced distortion can be generated. Although cubic convolution interpolation is computationally intensive and slow, the processing speed problem can be solved with hardware improvements. Hence, this study uses cubic convolution interpolation for image scaling.

C. GEI MULTIMODAL ATTENTION MECHANISM

Based on GaitSet, this study designs a channel attention mechanism that exploits the multimodal attention mechanism to focus attention on important areas of GEI and silhouette images to improve the gait recognition accuracy, as shown in FIGURE 6. Specifically, the attention mechanism employs a parameter-independent convolutional neural network to extract the feature representations of GEI and silhouette images. As a result, the key information in the feature map can



FIGURE 8. Gait contour diagram (a, NM state; b, BG state; c, CL state).

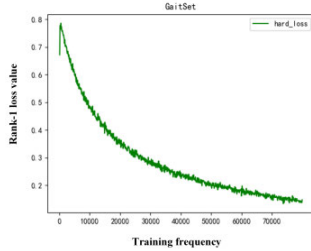


FIGURE 9. Loss function curve of the original model.

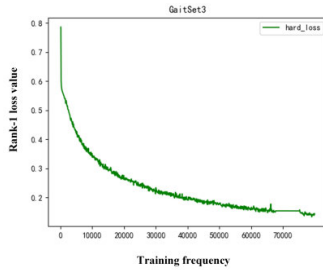


FIGURE 10. The loss function curve of the improved model.

be accurately screened out while excluding feature parameters that are not critical to the model performance [9]. The operation of each module includes global average pooling and global maximum pooling, merging the features after pooling, extracting deep features through the two fully connected layers, and obtaining the attention weight through sigmoid function normalization [13]. Weights accurately measure the relative importance of different regions, which are applied as key parameters in the weighted fusion of GEI and silhouette features.

The multimodal attention mechanism can be expressed as:

$$G = \alpha (f(s)) MPC [Maxpool(s), Avgpool(s)] \quad (6)$$

where α is the parameter generated after the convolution of the GEI and the silhouette image. Then, deep feature extraction is performed with the multimodal perception channel (MPC) after the global maximum pooling Max and the global average pooling Avg. The extraction results go through function $f(s)$ to obtain the final feature parameter G .

The equation for synthesizing a GEI is:

$$E_{GEI}(x, y) = \frac{1}{N} \sum_{n=1}^N G(x, y, n) \quad (7)$$

N is a complete gait cycle, $G(x, y, n)$ represents the n th frame of the silhouette image, and x and y represent the

horizontal and vertical coordinates of the silhouette, respectively. The GEI synthesis process is shown in FIGURE 7. The movement image of the human body in one cycle is normalized, and its energy is superimposed [22]. In this process, the pixel concentration actually reflects the movement frequency at that pixel position, i.e., the energy level.

This energy superposition not only reflects the dynamic information of human motion but also contains static information, thus providing us with a more comprehensive and in-depth description of motion characteristics. In this way, the attention is successfully focused on the important regions, which highlights the key feature representations of the two modes, thus significantly improving the model performance. In summary, the proposed method combining the multimodal attention mechanism with the channel attention mechanism yields excellent results in gait recognition tasks [23].

III. EXPERIMENT AND ANALYSIS

In this study, gait information was extracted from each video frame to obtain the binary contour of the human body. By using the normalized triple convolution interpolation method, different contour maps of pedestrians and their mean values were calculated. On this basis, the GEI was synthesized and used as an input feature to the neural network model. Moreover, this study proposed a multimodal attention mechanism that effectively extracts crucial information from the feature map. Finally, the results obtained in this study were validated using the CASIA-B database.

A. EXPERIMENTAL ENVIRONMENT

The hardware configuration is as follows: The computer is powered by an Intel Core i7-14700KF CPU with NVIDIA GeForce RTX 4080-16G graphics. The software environment consists of the Python 3.9 programming language and the Windows 11 operating system. For deep learning, the Pytorch 2.1 framework with CUDA version 12.1 is used for efficient computational acceleration. The overall configuration is stable and reliable to satisfy the requirements of deep learning.

B. DATABASE INTRODUCTION

The CASIA-B used in this study is currently recognized as a publicly available dataset in China, as shown in FIGURE 8. It includes gait data from 124 subjects, each at 11 different angles (0 to 180 degrees) for three different walking states (normal, NM; backpack, BG; and coat, CL). A categorical

TABLE 1. Comparison of recognition rates of NM, BG and CL Rank1 under ST division.

ST28		0°--180°											Mean
		0°	18°	36°	54°	72°	90°	108°	126°	144°	162°	180°	
NM	GaitSet	74.9	85.8	92.1	91.0	84.2	79.0	84.0	89.7	90.4	86.1	70.5	84.3
	Improved	74.6	85.1	93.1	91.1	83.8	80.0	84.4	91.0	90.4	85.8	70.2	84.5
BG	GaitSet	67.7	74.0	81.2	79.5	72.6	67.0	72.2	79.9	80.9	75.7	62.0	73.9
	Improved	65.4	77.8	82.5	81.1	74.8	70.3	73.7	80.5	80.7	76.1	62.7	75.1
CL	GaitSet	37.6	46.5	49.6	47.8	43.0	42.6	43.2	43.2	42.6	35.3	28.2	41.8
	Improved	37.0	48.5	49.8	47.5	45.2	44.2	46.2	46.3	42.2	36.7	28.5	42.9

TABLE 2. Comparison of recognition rates of NM, BG and CL Rank1 under MT division.

MT62		0°--180°											mean
		0°	18°	36°	54°	72°	90°	108°	126°	144°	162°	180°	
NM	GaitSet	88.5	95.1	97.3	96.3	89.9	88.2	91.3	96.2	97.5	95.0	83.7	92.6
	Improved	89.8	96.4	97.7	96.3	91.4	88.4	92.8	96.0	97.7	96.1	82.7	93.2
BG	GaitSet	78.8	87.5	92.2	89.8	84.4	78.3	82.3	89.2	92.0	88.5	76.5	85.4
	Improved	81.3	89.2	92.5	91.5	85.6	81.7	85.6	90.8	93.6	89.7	74.8	86.8
CL	GaitSet	58.6	68.8	73.3	66.9	60.1	57.7	61.8	66.7	68.6	64.2	51.1	63.4
	Improved	60.9	70.5	74.4	70.3	63.8	60.5	64.9	69.7	70.0	64.9	49.7	65.4

approach was used in this paper: The small sample training (ST) comprised a training set of 24 samples and a test set of 100 samples; the medium sample training (MT) included a training set of 62 samples and a test set of 62 samples; the large sample training (LT) consisted of a training set with 74 samples and a test set with 50 samples.

The training and test sets in this study were randomly sampled when $f=30$. The optimizer employed was Adam, which exhibited superior performance compared to Nadam in the Gaitset model. The learning rate was set to 1×10^{-4} . The loss function employed in this study was batch all + triplet loss (since hard loss did not converge, the full loss was used, which was the same loss function as GaitSet). The resolution of the camera was set to 320×240 ppi, and the resolution of the gait contour image, which was pre-processed by interpolation and other methods in Section II, was 64×64 ppi.

C. MAIN EXPERIMENTAL RESULTS

The Hard Loss curves of the loss function for both the original and improved models are depicted in FIGURE 9 and FIGURE 10, respectively. Upon comparing these figures, it can be seen that the improved model exhibits reduced

jitter compared to the original model, thereby indicating higher stability in this experimental setting. At the same time, it converges faster than the original model, which improves the recognition performance of the model. Additionally, the results of the original model demonstrate an upward trend in the initial segment with the occurrence of partial overfitting. However, no evidence of overfitting was observed in the experiment.

The GEI multimodal attention mechanism enables the model to dynamically adapt to its focus on various components during the processing of input data. This implies that the model can focus more on the information that is most important for gait feature extraction and recognition. Consequently, the model can efficiently leverage the relevant information from the input data, minimize redundant processing, and expedite the convergence process. By incorporating a multimodal attention mechanism, the model can precisely pinpoint gait contour features, thereby enhancing recognition accuracy.

TABLES 1, 2, and 3 show the experimental comparison results of ST, MT, and LT, respectively. It can be seen from the results that under the sample size of LT and different viewing angles, the recognition rate is 95.8% in the NM state, 89.3%

$$I_{new}(x, y) = \begin{bmatrix} u(x_1) \\ u(x_2) \\ u(x_3) \\ u(x_4) \end{bmatrix}^T \begin{bmatrix} I(x-x_1, y-y_1) & I(x-x_1, y-y_2) & I(x-x_1, y+y_3) & I(x-x_1, y+y_4) \\ I(x-x_2, y-y_1) & I(x-x_2, y-y_2) & I(x-x_2, y+y_3) & I(x-x_2, y+y_4) \\ I(x-x_3, y-y_1) & I(x-x_3, y-y_2) & I(x-x_3, y+y_3) & I(x-x_3, y+y_4) \\ I(x-x_4, y-y_1) & I(x-x_4, y-y_2) & I(x-x_4, y+y_3) & I(x-x_4, y+y_4) \end{bmatrix} \begin{bmatrix} u(y_1) \\ u(y_2) \\ u(y_3) \\ u(y_4) \end{bmatrix} \quad (5)$$

TABLE 3. Comparison of NM, BG, and CL Rank1 recognition rates under LT division.

LT74		0°--180°											mean
		0°	18°	36°	54°	72°	90°	108°	126°	144°	162°	180°	
NM	GaitSet	91.3	97.6	98.2	96.7	93.5	91.1	94.2	97.1	97.6	97.4	88.8	94.8
	Improved	92.9	98.0	99.3	97.8	93.7	91.8	95.6	98.0	99.2	97.8	89.1	95.8
BG	GaitSet	84.4	91.3	92.4	91.8	87.3	81.8	86.2	92.7	95.6	93.7	82.9	89.1
	Improved	87.0	82.8	93.1	92.0	85.6	81.9	86.3	92.7	95.2	93.8	82.3	89.3
CL	GaitSet	63.0	76.6	78.9	76.2	70.2	67.0	68.9	73.5	74.1	68.9	54.9	70.2
	Improved	69.5	76.6	80.0	76.8	72.7	68.5	72.7	75.2	75.5	71.6	58.4	72.5

TABLE 4. GEI multimodal attention mechanism ablation experiment.

Experiment	Improved normalized contour diagram	GEI attention mechanism	Rank1 recognition rate
1	no	no	42.4%
2	yes	no	85.7%
3	yes	yes	87.6%

in the BG state, and 72.5% in the CL state. Compared with the original Rank-1 model, the average accuracy is improved by 1.4%.

D. ABLATION EXPERIMENT

In order to evaluate the impact of the multimodal attention mechanism and interpolation method on overall performance, an ablation comparison experiment was conducted to gain deeper insights into the model's working principle, optimize its design, and enhance gait recognition accuracy.

The experimental conditions in Table 4 were consistent for the GaitSet model during experiments on the CASIA-B database. The experimental results represent the average values of ST, MT, and LT.

The ablation comparison experiment is described in TABLE 4, where the variables are the presence or absence of an improved normalized contour diagram and multimodal GEI attention fusion channel mechanisms under identical experimental conditions. It can be seen that the GEI multimodal attention channel mechanism and the improved normalized contour diagram proposed in this paper improve the performance of the GaitSet model.

IV. CONCLUSION

In this study, the image pre-processing is improved using cubic convolution interpolation, and the key feature representation is enhanced by a multimodal attention mechanism. The recognition performance of the model is significantly enhanced through these optimizations while maintaining the original pixel size. Notably, the recognition rate for individuals wearing a coat or jacket from the CL dataset is significantly improved, with an increasing rate of 3.3%. In addition, comparative experiments show that the improved model does not appear overfitting at the initial stage of

training, suggesting improved stability of the improved model.

In future research, we aim to conduct an in-depth analysis of the various factors that influence recognition rates. Our exploration will extend to the development of a multi-modal recognition approach, which synergistically combines gait recognition, pedestrian re-identification, and behavioral recognition, thereby expanding the applicability of this field. Furthermore, we intend to integrate motion capture technology to enhance image processing capabilities, meticulously accounting for the occlusion factor to refine our recognition methodologies.

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